The Discriminant-12 Return

From Centered Cayley Geometry to Pell, Cyclotomic, and Boolean Structure

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Abstract

Let $g \in SL_2$ have trace τ, centered part $H = g - \frac{\tau}{2}I$, and Cayley transform $K = (g-I)(g+I)^{-1}$. We prove the universal identities

$$K = \frac{2}{\tau+2}H, \qquad KH = \frac{\tau-2}{2}I.$$

Consequently, in the shear normalization $\tau = \sigma + 2$, the self-reciprocity condition $K = H^{-1}$ selects the unique value $\sigma = 2$. The period-$[1,2]$ primitive positive integral return is

$$g_{12} = \begin{pmatrix} 3 & 1 \\ 2 & 1 \end{pmatrix},$$

whose characteristic discriminant is 12 and whose eigenvalues are $2 \pm \sqrt{3}$. We show that its centered generator realizes multiplication by $\sqrt{3}$ on the ramified ideal $(2, \sqrt{3}-1)$, that the return preserves the primitive form $2x^2 - 2xy - y^2$, and that this form is a Lorentz norm in Pell coordinates. The same centered generator admits the cyclotomic completion $\mathcal{Z} = (H + iI)/2$, a primitive twelfth root whose integral closure carries a four-state parity quotient. Modulo 2, the return becomes factor exchange on a Boolean cell and its degenerate norm becomes XOR. The argument is algebraic and does not depend on zeta zeros, numerical certification, or the Riemann hypothesis.

1 Introduction

A hyperbolic matrix in SL_2 naturally produces two centered objects. The first is its traceless part

$$H = g - \frac{\mathrm{tr}(g)}{2}I.$$

The second is its Cayley transform

$$K = (g-I)(g+I)^{-1}.$$

Both lie on the one-dimensional traceless direction determined by g, but their scalar relation depends on the trace. Requiring them to be reciprocal turns out to be rigid: after the normalization $\mathrm{tr}(g) = \sigma + 2$, it forces

$$\sigma = 2.$$

This elementary selection rule is the entrance to a longer arithmetic chain. At the selected trace 4, a primitive positive integral representative has characteristic discriminant

$$\Delta_g = \mathrm{tr}(g)^2 - 4 = 12$$

and expanding eigenvalue $2+\sqrt{3}$. Its centered part squares to $3I$; its integral lattice is the ramified ideal above 2 in $\mathbb{Q}(\sqrt{3})$; its norm form is an integral Lorentz form; and adjoining the scalar complex structure produces a primitive twelfth root. Reduction at the ramified prime returns to a four-state Boolean cell, not as a translation but as factor exchange.

The purpose of this paper is to prove that chain in the order in which its dependencies actually occur:

$$\boxed{\begin{array}{c}\text{centered Cayley reciprocity} \Longrightarrow \sigma = 2 \Longrightarrow \Delta_g = 12 \Longrightarrow \mathbb{Q}(\sqrt{3}) \\ \Longrightarrow \text{Pell–Lorentz lattice} \Longrightarrow \mathbb{Q}(\zeta_{12}) \Longrightarrow \text{ramified Boolean factor exchange.}\end{array}}$$

Three distinctions will be maintained throughout. First, $\Delta_g = \operatorname{tr}(g)^2 - 4$ is the characteristic discriminant of a matrix; it is not automatically the discriminant of an arbitrarily chosen order. Second, the Lorentz dynamics is a discrete hyperbolic orbit, not a closed continuous orbit. Third, the Boolean translation group, the returned factor-exchange involution, and the cyclotomic Galois group are separate symmetry layers.

The algebraic theorem proved here is independent of any analytic realization. In particular, no identification with a Fredholm flow, no statement about zeta zeros, and no implication for the Riemann hypothesis is used or claimed.

2 Statement of the discriminant-12 return theorem

Theorem 2.1 (Discriminant-12 return)**.** *Let $g \in SL_2(\mathbb{Q})$, let $\tau = \operatorname{tr}(g)$, and assume $g+I$ is invertible. Define*

$$H = g - \frac{\tau}{2}I, \qquad K = (g-I)(g+I)^{-1}, \qquad \tau = \sigma + 2.$$

Then:

(i) $K = \dfrac{2}{\tau+2}H$ *and* $KH = \dfrac{\tau-2}{2}I.$

(ii) In the hyperbolic range, $K = H^{-1}$ if and only if $\sigma = 2$.

(iii) At $\sigma = 2$, the period-$[1,2]$ primitive positive integral representative

$$g_{12} = \begin{pmatrix} 3 & 1 \\ 2 & 1 \end{pmatrix}$$

has characteristic polynomial $X^2 - 4X + 1$, characteristic discriminant 12, and eigenvalues $2 \pm \sqrt{3}$. Its centered part

$$H_{12} = g_{12} - 2I = \begin{pmatrix} 1 & 1 \\ 2 & -1 \end{pmatrix}$$

satisfies $H_{12}^2 = 3I$.

(iv) On the lattice $\mathfrak{p}_2 = (2, \sqrt{3}-1)$, written in the ordered basis $(2, \sqrt{3}-1)$, the operators H_{12} and g_{12} are multiplication by $\sqrt{3}$ and $2+\sqrt{3}$, respectively. The associated primitive form

$$q_{12}(x,y) = 2x^2 - 2xy - y^2$$

is preserved by g_{12} and obeys

$$2q_{12}(x,y) = U^2 - V^2, \qquad (U,V) = (2x - y, \sqrt{3}\,y).$$

(v) *The element*

$$\mathcal{Z} = \frac{H_{12} + iI}{2}$$

satisfies $\Phi_{12}(\mathcal{Z}) = 0$ *and generates* $\mathbb{Q}(\zeta_{12})$*. Moreover*

$$g_{12} = -i(I + \mathcal{Z})(I - \mathcal{Z})^{-1}.$$

The maximal order is

$$\mathcal{O}_{\mathbb{Q}(\zeta_{12})} = \left\{ \frac{A + B\sqrt{3} + i(C + D\sqrt{3})}{2} : A \equiv D,\ B \equiv C \pmod 2 \right\},$$

and

$$\mathcal{O}_{\mathbb{Q}(\zeta_{12})}/\mathbb{Z}[\sqrt{3}, i] \cong (\mathbb{F}_2)^2.$$

(vi) *Modulo* 2*,*

$$\overline{g}_{12}(x, y) = (x + y, y), \qquad \overline{q}_{12}(x, y) = y.$$

Under $\Phi(x, y) = (\epsilon_1, \epsilon_2) = (x, x + y)$ *these become*

$$\Phi\overline{g}_{12}\Phi^{-1}(\epsilon_1, \epsilon_2) = (\epsilon_2, \epsilon_1), \qquad \overline{q}_{12} = \epsilon_1 \oplus \epsilon_2.$$

Thus the return is factor exchange and, together with Boolean translations, generates $V_4^{\mathrm{tr}} \rtimes C_2 \cong D_8$*.*

The theorem separates into four proof packages: universal centered-Cayley algebra, the integral Pell–Lorentz return, cyclotomic integral closure, and the ramified Boolean reduction. The first two are proved next.

3 Universal centered-Cayley algebra

The selection of the trace defect is a consequence of Cayley–Hamilton alone.

Lemma 3.1 (Centered square)**.** *For* $g \in SL_2$ *with trace* τ*,*

$$\boxed{H^2 = \left(\frac{\tau^2}{4} - 1\right) I.}$$

Proof. Cayley–Hamilton gives $g^2 - \tau g + I = 0$. Therefore

$$H^2 = \left(g - \frac{\tau}{2} I\right)^2 = g^2 - \tau g + \frac{\tau^2}{4} I = \left(\frac{\tau^2}{4} - 1\right) I.$$

□

Proposition 3.2 (Centered Cayley identity)**.** *If* $g + I$ *is invertible, then*

$$\boxed{K = \frac{2}{\tau + 2} H, \qquad KH = \frac{\tau - 2}{2} I.}$$

Proof. From Cayley–Hamilton,

$$(g+I)\big((\tau+1)I-g\big) = (\tau+2)I.$$

Hence

$$(g+I)^{-1} = \frac{(\tau+1)I-g}{\tau+2}.$$

Multiplication gives

$$\begin{aligned} K &= \frac{(g-I)((\tau+1)I-g)}{\tau+2} \\ &= \frac{(\tau+2)g-g^2-(\tau+1)I}{\tau+2} \\ &= \frac{2g-\tau I}{\tau+2} = \frac{2}{\tau+2}H. \end{aligned}$$

Using Lemma 3.1,

$$KH = \frac{2}{\tau+2}H^2 = \frac{2}{\tau+2}\left(\frac{\tau^2}{4}-1\right)I = \frac{\tau-2}{2}I.$$

□

Corollary 3.3 (Unique self-reciprocal class)**.** *Let $\tau = \sigma+2$ and assume g is hyperbolic. Then*

$$\boxed{K = H^{-1} \quad\Longleftrightarrow\quad \sigma = 2.}$$

Proof. Hyperbolicity implies H is invertible. By Proposition 3.2,

$$K = H^{-1} \quad\Longleftrightarrow\quad KH = I \quad\Longleftrightarrow\quad \frac{\tau-2}{2} = 1.$$

Thus $\tau = 4$, equivalently $\sigma = 2$. □

Remark 3.4. The number 2 is selected here before any integral representative, quadratic field, Pell equation, or analytic flow has been introduced. Later appearances of $2+\sqrt{3}$ and discriminant 12 are downstream consequences of this trace selection.

4 The primitive integral return

For the standard continued-fraction matrix

$$A_a = \begin{pmatrix} a & 1 \\ 1 & 0 \end{pmatrix},$$

the period $[1,2]$ gives, at the selected trace 4,

$$g_{12} = A_1A_2 = \begin{pmatrix} 3 & 1 \\ 2 & 1 \end{pmatrix}.$$

It is positive, primitive, and unimodular. Directly,

$$\det(g_{12}) = 1, \qquad \operatorname{tr}(g_{12}) = 4,$$

so

$$\chi_{g_{12}}(X) = X^2-4X+1, \qquad \Delta_{g_{12}} = 4^2-4 = 12.$$

The eigenvalues are $2\pm\sqrt{3}$.

Proposition 4.1 (Real quadratic generator)**.** *For*

$$H_{12} = g_{12} - 2I = \begin{pmatrix} 1 & 1 \\ 2 & -1 \end{pmatrix},$$

one has

$$H_{12}^2 = 3I, \qquad \mathbb{Z}[H_{12}] \cong \mathbb{Z}[\sqrt{3}] = \mathcal{O}_{\mathbb{Q}(\sqrt{3})}.$$

Proof. Matrix multiplication gives $H_{12}^2 = 3I$. Thus evaluation $\sqrt{3} \mapsto H_{12}$ defines an injective homomorphism $\mathbb{Z}[\sqrt{3}] \to M_2(\mathbb{Z})$ with image $\mathbb{Z}[H_{12}]$. Since $3 \equiv 3 \pmod 4$, the maximal order of $\mathbb{Q}(\sqrt{3})$ is $\mathbb{Z}[\sqrt{3}]$. □

Let

$$\mathfrak{p}_2 = (2, \sqrt{3} - 1).$$

This is the unique prime ideal above 2; indeed 2 ramifies in the real quadratic field of discriminant 12.

Proposition 4.2 (Ideal-lattice action)**.** *In the ordered basis*

$$e_1 = 2, \qquad e_2 = \sqrt{3} - 1$$

of $\mathfrak{p}_2$, multiplication by $\sqrt{3}$ and $2 + \sqrt{3}$ is represented by H_{12} and g_{12}, respectively.

Proof. The basis relations are

$$\sqrt{3}e_1 = e_1 + 2e_2, \qquad \sqrt{3}e_2 = e_1 - e_2.$$

Therefore the columns of the multiplication matrix are $(1, 2)^T$ and $(1, -1)^T$, giving H_{12}. Since multiplication by $2 + \sqrt{3}$ is $2I + H_{12}$, its matrix is g_{12}. □

5 The Pell–Lorentz norm form

The ideal norm becomes an integral indefinite binary quadratic form in the basis above. For

$$\alpha = xe_1 + ye_2 = (2x - y) + y\sqrt{3},$$

one has

$$N_{\mathbb{Q}(\sqrt{3})/\mathbb{Q}}(\alpha) = (2x - y)^2 - 3y^2 = 2\big(2x^2 - 2xy - y^2\big).$$

Define

$$q_{12}(x, y) = 2x^2 - 2xy - y^2.$$

Its binary-form discriminant is

$$(-2)^2 - 4(2)(-1) = 12.$$

Proposition 5.1 (Integral Lorentz return)**.** *The form q_{12} is primitive and is preserved by g_{12}. Under*

$$(U, V) = (2x - y, \sqrt{3}\,y),$$

it satisfies

$$\boxed{2q_{12}(x, y) = U^2 - V^2.}$$

Moreover g_{12} acts as the discrete Lorentz boost associated with the unit $2 + \sqrt{3}$.

Proof. Primitivity follows from the coefficients $2, -2, -1$. By Proposition 4.2, g_{12} is multiplication by the norm-one unit $2+\sqrt{3}$, so it preserves the ideal norm and hence q_{12}. The displayed coordinate change is the norm calculation above. In null coordinates $U \pm V$, multiplication by $2+\sqrt{3}$ scales the two eigendirections by $2 \pm \sqrt{3}$. Thus the iterates of g_{12} form a discrete hyperbolic Lorentz orbit on each level of q_{12}. □

Remark 5.2. The trace determines the recurrence law for the iterates, but not their initial data. Since g_{12} has characteristic polynomial $X^2 - 4X + 1$, every scalar coordinate of $g_{12}^n v$ obeys

$$a_{n+2} = 4a_{n+1} - a_n,$$

with initial values determined by v.

6 The geometric mod-12 unit shell

The arithmetic selected above also has a direct realization in the factor-cone coordinates of Paper A. The unit residues modulo 12 are

$$U(12) = (\mathbb{Z}/12\mathbb{Z})^\times = \{1, 5, 7, 11\}.$$

For each $r \in U(12)$, take the factor pair $(r, 1)$ together with its mirror $(1, r)$. Under

$$T = \frac{x+y}{2}, \qquad X = \frac{x-y}{2}, \qquad Y = \sqrt{xy},$$

the two points have

$$\boxed{T_r = \frac{r+1}{2}, \qquad X_r = \pm\frac{r-1}{2}, \qquad Y_r = \sqrt{r}, \qquad T_r^2 - X_r^2 = r.} \tag{1}$$

Thus the four unit labels give the exact data

r	1	5	7	11
T_r	1	3	4	6
$\lvert X_r\rvert$	0	2	3	5
Y_r	$\sqrt{1}$	$\sqrt{5}$	$\sqrt{7}$	$\sqrt{11}$.

The outer pair $r = 11$ is the factor pair $(11, 1)$ and its mirror $(1, 11)$; it therefore lies on the Paper-A anti-diagonal $x + y = 12$, or equivalently the fixed-radius cone circle $T = 6$. The remaining units occupy the nested circles $T = 1, 3, 4$ along the same distinguished row/column pair $x = 1$ and $y = 1$.

Proposition 6.1 (Mod-12 Klein group). *Every element of $U(12)$ has order dividing 2, and*

$$\boxed{U(12) \cong C_2 \times C_2 = V_4.}$$

Under the cyclotomic identification used below, $r \in U(12)$ acts by

$$\zeta_{12} \longmapsto \zeta_{12}^r,$$

so the same four labels index $\mathrm{Gal}(\mathbb{Q}(\zeta_{12})/\mathbb{Q})$.

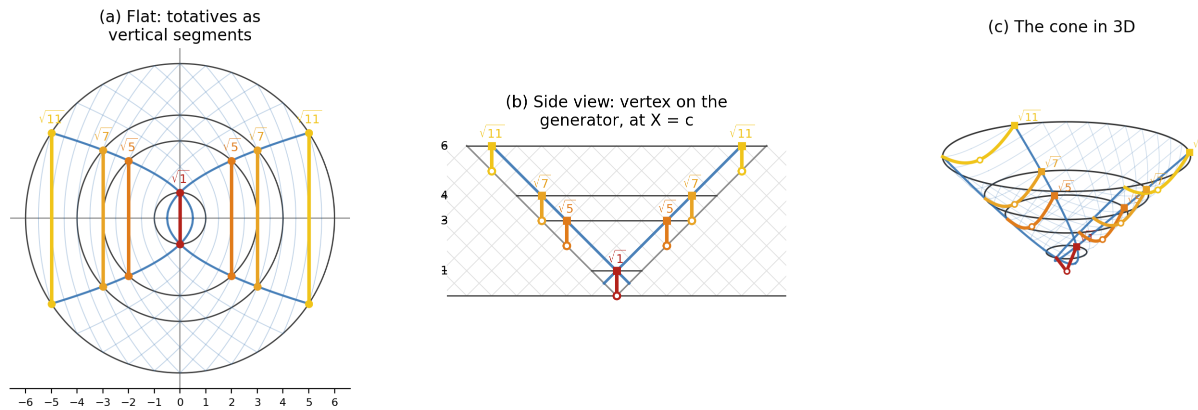

Figure 1: The mod-12 unit shell in factor-cone coordinates. For each $r \in \{1,5,7,11\} = U(12)$, the factor pairs $(r,1)$ and $(1,r)$ map to $T = (r+1)/2$, $X = \pm(r-1)/2$, and $Y = \sqrt{r}$, so that $T^2 - X^2 = r$. Panel (a) is the flat (X,Y) projection, panel (b) is the (X,T) side view, and panel (c) shows the corresponding curves on the null cone. The outer $r = 11$ pair lies on $x + y = 12$. The highlighted labels form V_4 under multiplication modulo 12; the cone embedding realizes their geometry but does not replace the modular group law.

Proof. Directly,

$$1^2 \equiv 5^2 \equiv 7^2 \equiv 11^2 \equiv 1 \pmod{12}.$$

The group has four elements and three nonidentity involutions, hence is $C_2 \times C_2$. The Galois statement is the standard cyclotomic identification $\mathrm{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q}) \cong (\mathbb{Z}/n\mathbb{Z})^\times$ at $n = 12$. □

Remark 6.2 (Layer boundary). Figure 1 is a geometric carrier for the four residue labels. Neither proximity nor adjacency in the drawing defines their multiplication. Moreover, the modulus 12, the characteristic discriminant $\Delta_{g_{12}} = 12$, the cyclotomic Galois V_4, and the Boolean translation V_4^{tr} remain logically distinct. Their agreement in the selected return is an exact compatibility developed in this paper, not a definitional identity.

7 Roadmap for the arithmetic completion

The remaining proof packages proceed without altering the hypotheses above. First, adjoining the scalar complex structure gives

$$\mathcal{Z} = \frac{H_{12} + iI}{2}, \qquad \mathcal{Z}\overline{\mathcal{Z}} = \frac{H_{12}^2 + I}{4} = I.$$

The half-element is therefore forced by relative-norm normalization and satisfies the twelfth cyclotomic polynomial. The integral closure introduces two parity-gluing classes, yielding a quotient isomorphic as an additive group to $(\mathbb{F}_2)^2$.

Second, reduction of the ramified ideal lattice modulo 2 gives another four-state space. The return reduces to the transvection

$$(x,y) \longmapsto (x+y, y),$$

which becomes factor exchange after the Boolean relabeling $(x,y) \mapsto (x, x+y)$. Its reduced norm becomes XOR. Equal cardinality does not canonically identify this ramified quotient with the cyclotomic gluing quotient, and neither should be identified with the cyclotomic Galois V_4.

These two packages complete parts (v) and (vi) of Theorem 2.1. They will be developed next from the integral parity description rather than inferred from the shared number of states.

Scope note

This draft establishes the theorem-first opening and the complete proofs of the centered-Cayley and Pell–Lorentz packages. A separate analytic companion may realize the selected value through a certified Suzuki determinant crossing. That realization is not a hypothesis of the algebraic theorem and is not used here.